

Representative Failure / Recovery Cases

Spatial recovery

Dataset: scanrefer Case: 1 Type: proximity_derived
Route: spatial-only text (pure_proximity_spatial)

Query: a counter / desk where a tv often sits . it is beside the black chair .

E0 IoU: 0.000 Routed IoU: 0.658
Transition: recovery

Spatial regression

Dataset: scanrefer Case: 54 Type: proximity_derived
Route: spatial-only text (pure_proximity_spatial)

Query: this picture is hanged on the wall besides another picture . two monitors are present on the table under these pictures .

E0 IoU: 0.376 Routed IoU: 0.000
Transition: regression

BEV recovery

Dataset: nr3d Case: 99 Type: ordinal
Route: BEV labeled layout (pure_ordinal_bev)

Query: facing the tables the second table from the left

E0 IoU: 0.000 Routed IoU: 1.000
Transition: recovery

Visual fallback kept correct

Dataset: scanrefer Case: 0 Type: explicit_direction
Route: E0 RGB canvas (visual_attribute_default_e0)

Query: the chair is west of the left - most table . the chair is dark brown and has four legs .

E0 IoU: 0.933 Routed IoU: 0.933
Transition: both_correct